Ga-Hyeon Eom

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RESEARCH PROFILE

Experimental researcher in self-sensing carbon nanotube (CNT)/polymer composites and printed piezoresistive sensors for safety helmets, electric-vehicle (EV) battery enclosures and concrete structures. Experience ranges from ink formulation and screen printing to sensor integration and impact/structural testing.

Research interests: Smart materials and structures; functional sensor and transducer fabrication; distributed sensing and sensor integration; mechanical interfacing of dissimilar materials and composites; scalable manufacturing by screen printing and extrusion-based 3D printing; impact and structural response sensing.

EDUCATION

M.S., Smart City Engineering (expected Feb. 2027)
Hanyang University ERICA, Ansan, South Korea **Mar. 2024–present**
GPA: 3.82 / 4.00 (cumulative)

B.S., Civil and Environmental Engineering
Hanyang University ERICA, Ansan, South Korea **Mar. 2019–Feb. 2024**

RESEARCH EXPERIENCE

Graduate Researcher
Department of Smart City Engineering, Hanyang University ERICA **Mar. 2024–present**

- **Impact-sensing array for construction safety helmets (first author).** Formulated screen-printable CNT/epoxy inks and printed piezoresistive sensors. Selected the CNT loading and sensor size from a nine-configuration drop-weight screening study. Integrated a 19-channel printed array (<1 g) inside a commercial helmet and co-developed the software for simultaneous multichannel readout. Three independently fabricated arrays detected all 240 crown impacts at 0.2–4.0 J, and off-crown impacts were localized with a median angular deviation of 6.2°. Published in Sensors (2026).
- **Battery safety monitoring (first author, IWSHM 2025).** Fabricated screen-printed CNT/epoxy patch sensors and related their conductivity to CNT loading and the number of print passes. On an EV battery enclosure, the sensors detected 0.2–0.8 J drop impacts, and a resistance-change map localized the impact point. Co-inventor on patent applications for enclosure-embedded impact and flame detection and for an inter-cell battery sensor. Now developing multichannel CNT/epoxy temperature-sensor sheets for battery thermal monitoring.
- **Self-sensing CNT composites for concrete structures (first author).** Fabricated CNT-reinforced polymer (CNRP) bars and CNT/epoxy coatings and tested them in concrete beams under three-point bending; both detected crack initiation as a sharp rise in resistance. Participated in domestic bridge field tests of scalable CNT patch sensors (sensor fabrication, field testing and data analysis) and in a pre-deployment site survey of a bridge in Baltimore, MD, with project partner AECOM (2025).
- **Other contributions.** Worked on CNT/epoxy conductive repair materials (DRIFT, a EUREKA Network project) and screen-printed CNT/polyurethane temperature sensors. Co-authored studies on extrusion-based 3D printing of CNT/polyurethane sensors (CAMX 2024), on the environmental durability of CNT sensors and on induction-heated de-icing concrete (with Dalian University of Technology).

SKILLS AND LANGUAGES

Materials: MWCNT dispersion in epoxy and polyurethane (probe sonication, three-roll milling); screen-printable CNT/epoxy inks; CNT/epoxy coatings and repair materials for concrete

Fabrication: Screen printing of piezoresistive and temperature sensors; silver-paint electrodes; molded CNRP bars; sensor integration on curved helmet shells, battery enclosures and concrete beams

Testing: Resistance and conductivity measurement (Keithley 2450); percolation analysis; drop-weight impact tests; three-point bending with in-situ resistance monitoring; temperature/humidity exposure

Data: Multichannel resistance acquisition and threshold-based event detection (co-developed); data analysis and plotting (OriginPro, Python scripting)

Languages: Korean (native); English (8 English-taught graduate courses, 24 credits)

PUBLICATIONS

*Bold = author of this CV. * = corresponding author.*

Journal Articles and Manuscripts

1. **G.-H. Eom**, Y.-J. Jung, P. Sagar, S.-H. Jang*, "A Screen-Printed Carbon Nanotube/Epoxy Array for Impact Monitoring in Construction Safety Helmets", Sensors, vol. 26, no. 20, art. 6557, 2026, doi:10.3390/s26206557.
2. X. Ouyang, S. Dong*, Y. Xiong, Z. Li, **G.-H. Eom**, S.-H. Jang, B. Han*, "Optimizing Electromagnetic Induction Heating and Rapid De-Icing Properties of Ultra-High Performance Concrete: From Multiscale Fillers to Induction Condition", Applied Thermal Engineering, submitted Sep. 2026.
3. T. Ban, S. Yun, S. Kim, J. Park, W. Jeong, J. Park, J. Park, H. Park, **G.-H. Eom**, S.-J. Lee, S.-H. Jang, N. Ahmadi, B. J. Park, J.-M. Kim*, "Structurally Gated Multilayer Hybrid Architecture for Programmable Mechanochromic Sensing", Advanced Composites and Hybrid Materials, revision submitted Sep. 2026.

Conference Proceedings (Full Papers)

1. Y.-J. Jung, **G.-H. Eom**, H.-J. Heo, S.-H. Jang, "Development of a Rapid-Curing CNT-Epoxy Composite with Recycled Concrete Aggregate for Structural Repair", 17th International Symposium on Fiber-Reinforced Polymer (FRP) Reinforcement for Concrete Structures (FRPRCS17), Girona, Spain, 2026 (proceedings forthcoming).
2. **G.-H. Eom**, Y.-J. Jung, S. Sen, S.-H. Jang*, "Screen-Printed Carbon Nanotube Polymer Composites for Impact Sensing in Electric Vehicle Batteries", 15th International Workshop on Structural Health Monitoring (IWSHM 2025), Stanford, CA, USA, pp. 1612–1618, 2025, doi:10.12783/shm2025/37463.
3. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, "Performance Evaluation of Structural Behavior Monitoring Using Smart Sensors", 24th International Conference on Composite Materials (ICCM24), Baltimore, MD, USA, 2025, doi:10.5281/zenodo.18597829.
4. H.-K. Jeon, Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, S. Sen, "Characterization of the 3D-Printed Carbon Nanotube/Polyurethane Composite with Structural Health Monitoring Capabilities", Composites and Advanced Materials Expo (CAMX 2024), San Diego, CA, USA, 2024, doi:10.33599/nasampe/c.24.0284.
5. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, "Characterization of Carbon Nanotube-Polyurethane Sheet Using Screen Printing Technology for Compression and Impact Sensing", Society for the Advancement of Material and Process Engineering 2024 (SAMPE 2024), Long Beach, CA, USA, 2024, doi:10.33599/nasampe/s.24.0072.

PATENTS (SOUTH KOREA)

Registered

1. S.-H. Jang, S.-J. Lee, J.-M. Kim, D.-G. Lee, C.-B. Kim, D.-W. Jang, **G.-H. Eom**, "Deformation detection device for tunnel lining concrete structures", Patent No. 10-3040533, registered Oct. 26, 2026.
2. S.-H. Jang, Y.-J. Jung, **G.-H. Eom**, "Reinforcing bar based on carbon nanotube, apparatus and system of monitoring of structure using the same and method thereof", Patent No. 10-2959099, registered Apr. 27, 2026.
3. S.-H. Jang, Y.-J. Jung, **G.-H. Eom**, "Concrete internal temperature detection system using carbon nanotube sensor", Patent No. 10-2959485, registered Apr. 27, 2026.
4. S.-H. Jang, Y.-J. Jung, **G.-H. Eom**, "Concrete load detection system utilizing carbon nanotube sensor", Patent No. 10-2909450, registered Jan. 5, 2026.
5. S.-H. Jang, Y.-J. Jung, **G.-H. Eom**, "Carbon nanotube-based wood adhesive composition and manufacturing method thereof", Patent No. 10-2881472, registered Oct. 31, 2025.
6. S.-H. Jang, H.-K. Jeon, Y.-J. Jung, **G.-H. Eom**, "Bag including carbon nanotube sheet and method for measuring load of inside of the bag using the same", Patent No. 10-2849694, registered Aug. 20, 2025.

Pending Applications

1. S.-H. Jang, **G.-H. Eom**, Y.-J. Jung, "Battery cell fixing bracket having temperature sensing function and battery pack including the same", Application No. 10-2026-0151526, filed Aug. 13, 2026.
2. S.-H. Jang, **G.-H. Eom**, Y.-J. Jung, "Roof snow removal panel and roof snow removal system having the same", Application No. 10-2026-0080950, filed May 6, 2026.

3. S.-H. Jang, **G.-H. Eom**, Y.-J. Jung, "Inter-cell insertion type battery monitoring sensor, method for manufacturing the same, and battery management system using the same", Application No. 10-2026-0057536, filed Mar. 31, 2026.
4. S.-H. Jang, Y.-J. Jung, **G.-H. Eom**, "Impact detection system and method for manufacturing sensors embedded in carbon nanotube-based battery enclosure", Application No. 10-2025-0005589, filed Jan. 14, 2025.
5. S.-H. Jang, Y.-J. Jung, **G.-H. Eom**, "Flame detection system and method for manufacturing sensors embedded in carbon nanotube-based battery enclosure", Application No. 10-2025-0005588, filed Jan. 14, 2025.
6. S.-H. Jang, **G.-H. Eom**, Y.-J. Jung, H.-K. Jeon, "Method of detecting for deformation regarding carbon nanotube reinforcement bar or structure using the same and system thereof", Application No. 10-2024-0017253, filed Feb. 5, 2024.

HONORS AND AWARDS

2026 Excellent Paper Award (co-author), Korean Society for Advanced Composite Structures (KOSACS) Conference

2025 Outstanding Performance Award (top tier), BK21 Smart City Research Group, Hanyang University

2025 Excellent Paper Award (co-author), Korean Society for Advanced Composite Structures (KOSACS) Conference

2025 Excellent Paper Presentation Award (first author, Special Session), Korea Institute for Structural Maintenance and Inspection (KSMI) Spring Conference

2024 Outstanding Performance Award (top tier), BK21 Smart City Research Group, Hanyang University

2024 Excellence Award (PI), Women Graduate Student Engineering Research Team Support Project presentation competition, Korean Society of Civil Engineers (KSCE)

2024 Excellent Paper Presentation Award (co-author), Korea Institute for Structural Maintenance and Inspection (KSMI) Fall Conference

2024 Excellent Paper Award (co-author), Korean Society for Advanced Composite Structures (KOSACS) Conference

2024 Excellent Paper Presentation Award (first author), Korea Institute for Structural Maintenance and Inspection (KSMI) Spring Conference

FUNDED RESEARCH PROJECTS

Dates shown are official project periods.

DRIFT: Disaster Reconnaissance & Intelligent Forecasting Technology
Ministry of Trade, Industry and Energy (EUREKA Network project) | *Student Researcher* **Jul. 2025–Jun. 2027**
• CNT/epoxy composite-based conductive repair materials for resilient infrastructure monitoring.

Development of a Flexible Modular AI-based Battery and Monitoring System for Urban Transportation
Korea Agency for Infrastructure Technology Advancement (KAIA) | *Project Coordinator* **Jul. 2024–Mar. 2027**
• Korea–UK consortium; printed CNT/epoxy impact sensors for battery enclosures, inter-cell monitoring sensor sheets and thermal sensor sheets.

Development of a Carbon Nanotube-Based Self-Sensing Rod for Safety Monitoring of Concrete Structures
Korea Foundation for Women in Science, Engineering and Technology (WISET) | *PI* **Apr. 2024–Oct. 2024**
• Student research team grant (2024 Women Graduate Student Engineering Research Team Support Project), 8,000,000 KRW. Led a team of three undergraduates on CNRP bars and CNT/epoxy coatings for concrete; the team received the Excellence Award at the program's 2024 presentation competition (Korean Society of Civil Engineers).

Structural Health Monitoring System using Scalable Carbon Nanotube Sensors
Korea Agency for Infrastructure Technology Advancement (KAIA) | *Student Researcher* **Jun. 2023–Dec. 2027**
• Korea–US project with AECOM; domestic bridge field tests of scalable CNT patch sensors.

Development of a Smart Monitoring System Integrating 3D Printed Battery-Free Antenna Sensor Technology with AI Optimization
Ministry of Trade, Industry and Energy | *Student Researcher* **Apr. 2021–Dec. 2024**
• Contributed to studies on extrusion-based 3D printing of CNT/polyurethane composite sensors (co-author, CAMX 2024).

Development of Intelligent Construction Materials System for Multi-Sensing and Self-Heating from Extreme Environments

Ministry of Science and ICT | Student Researcher **Mar. 2020–Feb. 2024**

• CNT/polyurethane and CNT/epoxy composites for IoT-based structural monitoring and extreme-environment applications.

INTERNATIONAL RESEARCH EXPERIENCE AND EXHIBITION

CES 2026, Las Vegas, NV, USA

Exhibitor team member, Hanyang University pavilion, Eureka Park (Venetian Expo) **Jan. 2026**

• Staffed the lab's booth presenting its CNT/polymer sensor technologies, including printed battery-monitoring sensors.

Overseas Research Training, Graduate School Innovation Support Program, Hanyang University ERICA

Baltimore, MD and New York, NY, USA **Aug. 2025**

• Presented at ICCM24 (Baltimore); took part in a technical meeting and a bridge site survey with AECOM for CNT patch-sensor deployment; visited Prof. Huiming Yin's group at Columbia University.

CONFERENCE CONTRIBUTIONS

International Conferences

1. Y.-J. Jung, **G.-H. Eom**, H.-J. Heo, S.-H. Jang, "Development of a Rapid-Curing CNT-Epoxy Composite with Recycled Concrete Aggregate for Structural Repair", 17th International Symposium on Fiber-Reinforced Polymer (FRP) Reinforcement for Concrete Structures (FRPRCS17), Girona, Spain, 2026.
2. **G.-H. Eom**, Y.-J. Jung, S.-H. Jang, "Embedded CNRP Bars for Real-Time Structural Health Monitoring of Concrete Structures", Engineering Mechanics Institute Conference 2026 (EMI 2026), Boulder, CO, USA, 2026.
3. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, "Self-Sensing CNT/Epoxy Concrete for Rapid Pothole Repair and Post Repair Monitoring", Engineering Mechanics Institute Conference 2026 (EMI 2026), Boulder, CO, USA, 2026.
4. **G.-H. Eom**, Y.-J. Jung, S.-H. Jang, "Performance Evaluation of Structural Behavior Monitoring Using Smart Sensors", 24th International Conference on Composite Materials (ICCM24), Baltimore, MD, USA, 2025.
5. **G.-H. Eom**, Y.-J. Jung, S.-H. Jang, "Safety Monitoring of Concrete Structures Using Carbon Nanotube Reinforced Polymer Bars", Engineering Mechanics Institute Conference 2025 (EMI 2025), Anaheim, CA, USA, 2025.
6. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, "Development of Conductive Repair Materials with Self-Sensing Functionality for Pothole Maintenance", Engineering Mechanics Institute Conference 2025 (EMI 2025), Anaheim, CA, USA, 2025.
7. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, "Evaluating Carbon Nanotube Smart Sensors for Crack Monitoring in Concrete Structures: Bridge Field Test", American Concrete Institute (ACI) Concrete Convention – Fall 2024, Philadelphia, PA, USA, 2024.
8. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, "Self-Sensing Cementitious Composite Using Carbon Nanotubes", Engineering Mechanics Institute and Probabilistic Mechanics & Reliability Conference 2024 (EMI/PMC 2024), Chicago, IL, USA, 2024.
9. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, "Characterization of Carbon Nanotube-Polyurethane Sheet Using Screen Printing Technology for Compression and Impact Sensing", Society for the Advancement of Material and Process Engineering 2024 (SAMPE 2024), Long Beach, CA, USA, 2024.

Domestic Conferences (South Korea)

1. **G.-H. Eom**, Y.-J. Jung, L.-C. Liu, S.-H. Jang, "A Screen-Printed CNT/Epoxy Piezoresistive Sensor Array for Low-Energy Impact Monitoring in Construction Safety Helmets", Korea Institute for Structural Maintenance and Inspection 2026 Fall Conference (KSMI 2026), Jeju, South Korea, 2026.
2. **G.-H. Eom**, Y.-J. Jung, D.-H. Kim, L.-C. Liu, S.-H. Jang, "Development of a Screen-Printed CNT/Epoxy Composite Temperature Sensor Sheet for Battery Thermal Monitoring", Korean Society for Advanced Composite Structures 2026 Conference (KOSACS 2026), Jeju, South Korea, 2026.
3. Y.-J. Jung, **G.-H. Eom**, H.-J. Heo, S.-H. Jang, "Development of a Carbon Nanotube Based Sensor for Monitoring Concrete Filling State", Korean Society for Advanced Composite Structures 2026 Conference (KOSACS 2026), Jeju, South Korea, 2026 [Excellent Paper Award].

4. H.-J. Heo, Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Development of a Rapid Curing Self-Sensing Repair Material Based on CNT/Epoxy Composites”, Korean Society for Advanced Composite Structures 2026 Conference (KOSACS 2026), Jeju, South Korea, 2026.
5. **G.-H. Eom**, Y.-J. Jung, D.-H. Kim, L.-C. Liu, S.-H. Jang, “Development and Performance Evaluation of a Carbon Nanotube Composite-Based Embedded Impact Sensor for Safety Helmets”, Korea Institute for Structural Maintenance and Inspection 2026 Spring Conference (KSMI 2026), Busan, South Korea, 2026.
6. Y.-J. Jung, **G.-H. Eom**, H.-J. Heo, S.-H. Jang, “Development of a Self-Sensing Rapid Repair Material Based on Carbon Nanotube Polymer Composites”, Korea Institute for Structural Maintenance and Inspection 2026 Spring Conference (KSMI 2026), Busan, South Korea, 2026.
7. L.-C. Liu, Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “A Multi-Parameter Numerical Study on Concrete Void Detection Using CNT Piezoresistive Sensors”, Korea Institute for Structural Maintenance and Inspection 2026 Spring Conference (KSMI 2026), Busan, South Korea, 2026.
8. S.-J. Lee, **G.-H. Eom**, Y.-K. Kim, B.-W. Han, S.-H. Jang, “Field Monitoring and Performance Evaluation of Levees Using Smart Sensors”, Korea Institute for Structural Maintenance and Inspection 2026 Spring Conference (KSMI 2026), Busan, South Korea, 2026.
9. **G.-H. Eom**, Y.-J. Jung, D.-H. Kim, H.-M. Yoo, S.-H. Jang, “Development of Carbon Nanotube-Based Sensors for Real-Time Impact Monitoring in Smart Helmets”, Korean Society for Composite Materials 2025 Fall Conference (KSCM 2025), Seoul, South Korea, 2025.
10. Y.-J. Jung, **G.-H. Eom**, M.-S. Cho, H.-J. Heo, S.-H. Jang, “Carbon Nanotube Composite-Based Formwork-Attached Sensor for Concrete Casting Monitoring”, Korean Society for Composite Materials 2025 Fall Conference (KSCM 2025), Seoul, South Korea, 2025.
11. **G.-H. Eom**, Y.-J. Jung, D.-H. Kim, H.-M. Yoo, S.-H. Jang, “Development of a Carbon Nanotube-Based Sensor for Helmet Impact Monitoring”, Korea Institute for Structural Maintenance and Inspection 2025 Fall Conference (KSMI 2025), Gwangju, South Korea, 2025.
12. Y.-J. Jung, **G.-H. Eom**, H.-J. Heo, M.-S. Cho, S.-H. Jang, “Monitoring of Concrete Placement Using Carbon Nanotube/Polymer Composite-Based Sensors”, Korea Institute for Structural Maintenance and Inspection 2025 Fall Conference (KSMI 2025), Gwangju, South Korea, 2025.
13. S.-J. Lee, **G.-H. Eom**, Y.-K. Kim, B.-W. Han, S.-H. Jang, “Development and Performance Analysis of a Levee Monitoring Sensor Using Carbon Nanotubes”, Korea Institute for Structural Maintenance and Inspection 2025 Fall Conference (KSMI 2025), Gwangju, South Korea, 2025.
14. **G.-H. Eom**, Y.-J. Jung, S.-H. Jang, “Development of a Carbon Nanotube-Based Impact Sensor for Battery Safety Monitoring”, Korean Society for Composite Materials 2025 Spring Conference (KSCM 2025), Jeju, South Korea, 2025.
15. **G.-H. Eom**, Y.-J. Jung, K.-M. Kim, J.-S. Oh, S.-H. Jang, “Safety Monitoring of Concrete Structures Using Carbon Nanotube Reinforced Polymer Bars”, Korean Society for Advanced Composite Structures 2025 Conference (KOSACS 2025), Jeonju, South Korea, 2025.
16. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Environmental Durability Analysis and Monitoring Performance Evaluation of Smart Sensors”, Korean Society for Advanced Composite Structures 2025 Conference (KOSACS 2025), Jeonju, South Korea, 2025 [Excellent Paper Award].
17. **G.-H. Eom**, Y.-J. Jung, K.-M. Kim, S.-H. Jang, “Development of Embedded Carbon Nanotube-Based Sensors for Structural Health Monitoring”, Korea Institute for Structural Maintenance and Inspection 2025 Spring Conference (KSMI 2025), Jeju, South Korea, 2025 [Excellent Paper Presentation Award].
18. Y.-J. Jung, **G.-H. Eom**, J.-Y. Kim, J.-M. Park, S.-H. Jang, “Evaluation of Ultraviolet Aging Effects on Carbon Nanotube Polymer Composite Sensors”, Korea Institute for Structural Maintenance and Inspection 2025 Spring Conference (KSMI 2025), Jeju, South Korea, 2025.
19. S.-J. Lee, **G.-H. Eom**, Y.-K. Kim, B.-W. Han, S.-H. Jang, “Development of a Levee Monitoring System Using Smart Sensors”, Korea Institute for Structural Maintenance and Inspection 2025 Spring Conference (KSMI 2025), Jeju, South Korea, 2025.
20. **G.-H. Eom**, Y.-J. Jung, S.-H. Jang, “Development of Smart Repair Materials for Structural Health Monitoring”, Korea Concrete Institute 2024 Fall Conference (KCI 2024), Goseong, South Korea, 2024.
21. **G.-H. Eom**, Y.-J. Jung, S.-H. Park, J.-Y. Kim, S.-H. Jang, “Development of Carbon Nanotube/Epoxy Coating on Concrete for Structural Health Monitoring”, Korean Society of Civil Engineers 2024 Convention (KSCE 2024), Jeju, South Korea, 2024.

22. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Detection of Concrete Structure Damage Using Carbon Nanotube-Based Smart Sensors”, Korean Society of Civil Engineers 2024 Convention (KSCE 2024), Jeju, South Korea, 2024.
23. **G.-H. Eom**, Y.-J. Jung, S.-H. Jang, “Development of Carbon Nanotube-Based Temperature Sensors Using Screen Printing”, Korea Institute for Structural Maintenance and Inspection 2024 Fall Conference (KSMI 2024), Busan, South Korea, 2024.
24. Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Analysis of Sensing Characteristics of CNT/PU Composites under Multiple Physical Stimuli”, Korea Institute for Structural Maintenance and Inspection 2024 Fall Conference (KSMI 2024), Busan, South Korea, 2024 [Excellent Paper Presentation Award].
25. **G.-H. Eom**, Y.-J. Jung, H.-K. Jeon, S.-H. Jang, “Development of a Concrete-Embedded Sensor Using Carbon Nanotubes”, Korean Society for Advanced Composite Structures 2024 Conference (KOSACS 2024), Jeju, South Korea, 2024.
26. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, “Development of a Load Sensing System Based on Carbon Nanotube/Polyurethane Composites”, Korean Society for Advanced Composite Structures 2024 Conference (KOSACS 2024), Jeju, South Korea, 2024 [Excellent Paper Award].
27. H.-K. Jeon, Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Behavior Evaluation of Structural Repair Areas Using Polymer Concrete Based on Carbon Nanotubes”, Korean Society for Advanced Composite Structures 2024 Conference (KOSACS 2024), Jeju, South Korea, 2024.
28. **G.-H. Eom**, Y.-J. Jung, H.-K. Jeon, S.-H. Jang, “Behavioral Evaluation of Concrete Structures Using CNRP Bar”, Korea Institute for Structural Maintenance and Inspection 2024 Spring Conference (KSMI 2024), Jeju, South Korea, 2024 [Excellent Paper Presentation Award].
29. H.-K. Jeon, Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Performance Evaluation of Smart Repair Materials for Monitoring Repair Areas”, Korea Institute for Structural Maintenance and Inspection 2024 Spring Conference (KSMI 2024), Jeju, South Korea, 2024.
30. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, “Development of Smart Monitoring System Utilizing Carbon Nanotube/Polyurethane Composites”, Korea Institute for Structural Maintenance and Inspection 2024 Spring Conference (KSMI 2024), Jeju, South Korea, 2024.
31. S.-J. Lee, Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, “Characteristics of the Ice Detection and Removal System Based on Carbon Nanotubes”, Korea Institute for Structural Maintenance and Inspection 2023 Fall Conference (KSMI 2023), Yangyang, South Korea, 2023.
32. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, “Development of Electronic Skin and Its Application to Concrete Structural Load Monitoring”, Korea Concrete Institute 2022 Spring Conference (KCI 2022), Jeju, South Korea, 2022.
33. S.-J. Lee, Y.-J. Jung, **G.-H. Eom**, S.-H. Jang, “Development of a Carbon Nanotube-Based Composite for Temperature Sensing of Mechanical Structures”, Korea Institute for Structural Maintenance and Inspection 2022 Spring Conference (KSMI 2022), Gwangju, South Korea, 2022.
34. Y.-J. Jung, H.-K. Jeon, **G.-H. Eom**, S.-H. Jang, “Performance Evaluation of E-Skin for Structural Deformation Detection”, Korean Society of Hazard Mitigation 2022 Conference (KOSHAM 2022), Seoul, South Korea, 2022 [Excellent Paper Award, poster session].